

Biacore T200

Biacore T200 (Fig 1) is a versatile, label-free system for detailed studies of biomolecular interactions, from early research to drug discovery and development, and on to QC.

The system delivers high quality kinetic, affinity, concentration, specificity, selectivity, and thermodynamic interaction data – in real time with exceptional sensitivity. Interactions characterized by on- and off-rates at the extremities of the kinetic scale can be analyzed with greater precision and confidence than ever before.

- Obtain high quality kinetics from the fastest on-rates to the slowest off-rates
- Analyze interactions involving the smallest low molecular weight (LMW) compounds
- Process up to 384 samples in unattended runs
- Get to the final result faster using guided workflows with built-in data quality assessments and tools, such as single cycle kinetics and calibration-free concentration analysis
- Recover samples for identification by mass spectrometry
- Apply dedicated tools for confident immunogenicity testing

Biacore T200 features wizard-supported software for assay development, analysis, data evaluation, and interpretation. Alternatively, methods may be developed or customized as required. In addition to providing detailed information on kinetics and affinity, software support allows interactions to be thermodynamically characterized.

An optional GxP package supports Biacore T200 operation in compliance with regulatory demands.



Fig 1. Biacore T200 for high sensitivity, label-free interaction analysis.

Main applications:

The performance and versatility of Biacore T200 provides several advantages for a large range of applications involving biomolecular interaction studies.

- Increase understanding of molecular mechanisms and structure-function relationships
- Define potential drug targets and diagnostic markers
- Select and characterize biotherapeutic candidates
- Select and optimize lead compounds during drug discovery
- Detect and characterize anti-drug antibodies (ADA) in immunogenicity studies
- Perform time- and cost-efficient concentration analysis in vaccine development



Pushing detection limits of label-free interaction analysis

The sensitivity of Biacore T200 enables the precise affinity analysis of the smallest organic compounds (Fig 2) and extends the range of kinetic values that can be precisely determined. Previously borderline data may thus be confidently interpreted, enabling kinetic analysis of the simplest analytes, as well as detection of low abundance proteins.

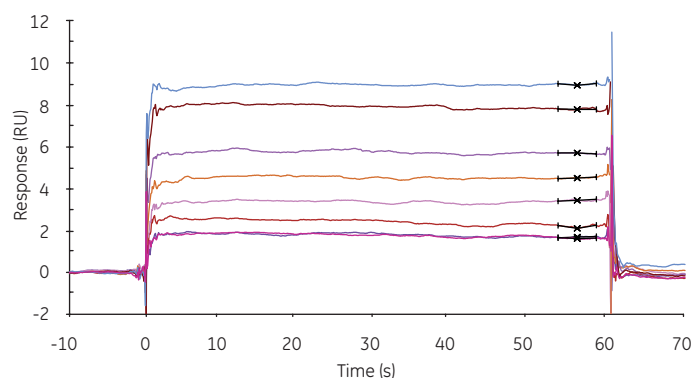


Fig 2. Interaction profile of methanesulfonamide (M, 95) with carbonic anhydrase.

Biacore T200 is designed for the analysis of molecular interactions where high sensitivity is crucial:

- Small organic compounds (no minimum MW limit)
- Low abundance molecules (concentration > 10 pM)
- Rare or sensitive targets such as G protein-coupled receptors
- Avoidance of avidity effects when analyzing interactions with bivalent antibodies
- Weak interactions, K_D in mM range
- Stable binders, $k_d \geq 10^{-5} \text{ s}^{-1}$

Working with rare or sensitive targets

The possibility to derive high quality data from low levels of immobilized proteins is advantageous in the analysis of interactions between small molecules and sensitive proteins such as G protein-coupled receptors, which are among the most important drug targets. The high sensitivity of Biacore T200 means that preservation of function is necessary in only a fraction of the total immobilized GPCRs (Fig 3). In addition, rare targets may be used sparingly, allowing for reduced consumption with no risk of compromise in data quality. Sensitive targets may thus be studied with greater confidence, reducing time-to-results in the drug discovery process.

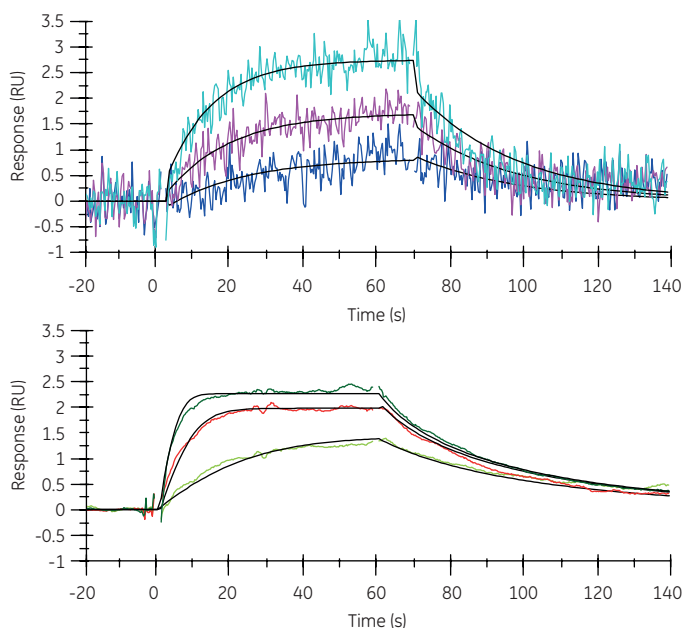


Fig 3. Binding of a small molecule, xanthine amine congener (XAC), to stabilized histidine-tagged GPCR (StaR®) A2. Data of higher quality is generated by Biacore T200 (lower sensorgram) compared to Biacore T100 (upper sensorgram) when using low levels of immobilized GPCR. Data courtesy of Dr. Andrei Zhukov, Heptares Therapeutics Ltd, Welwyn Garden City, UK.

Increased flexibility in assay design

With Biacore assays, it is usually preferable to immobilize antibody on the sensor surface, rather than in solution, in order to avoid complicating effects of avidity arising when antibodies attach to a surface densely coated with antigen. The sensitivity of Biacore T200 is sufficient to allow antigen immobilization levels so low that avidity is avoided (Fig 4), thus adding full flexibility to assay design.

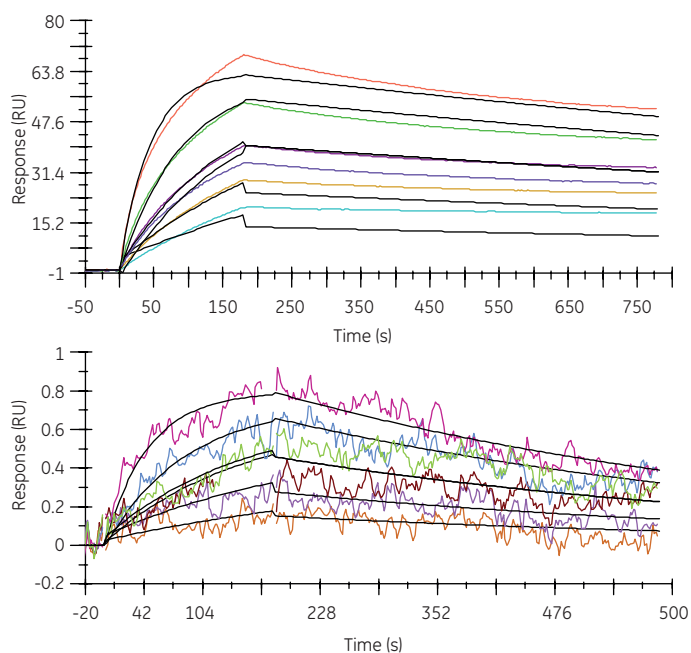


Fig 4. At low immobilization levels of antigen (lower sensorgrams), interaction data are better fitted to a 1:1 model. Kinetic rate constants of interactions involving antibodies as the binding partner in solution are thus calculated with greater precision and confidence.

High precision over the broadest kinetic range

The microfluidic system in Biacore T200 is optimized for the highest quality kinetics. The four flow cells can be used for single, paired, or serial runs. Paired, on-chip flow cell connections lead to minimum void volume between flow cells, ensuring accurate reference subtraction.

Biacore T200 enables measurement of kinetic constants over the broadest range, from the fastest on-rates to the slowest off-rates (Fig 5). This enables confident ranking of strong binders, which can be important in antibody selection. It also enables the detection of differences among rapid binders, an important feature when studying biological processes limited by bioavailability.

- k_{on} from 10^3 to $5 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$ (10^3 to $3 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ for macromolecular analytes)
- k_{d} from 10^{-5} to 1 s^{-1}

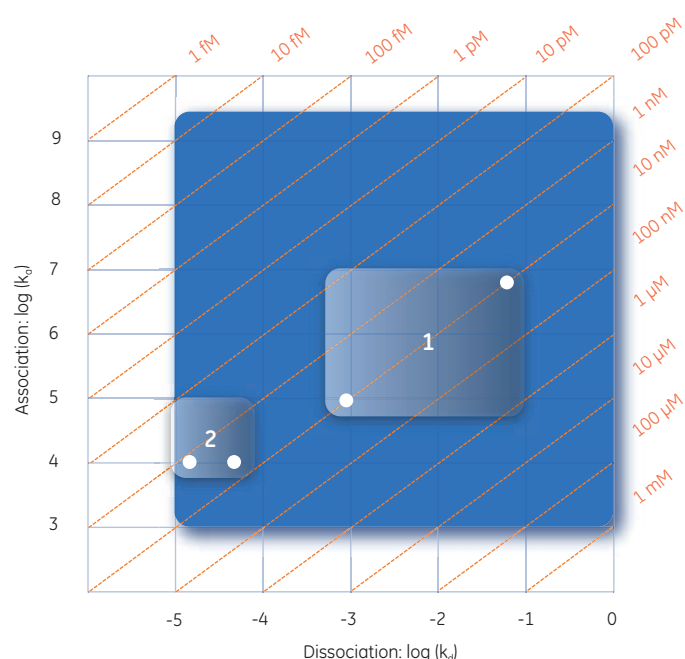


Fig 5. Kinetic measurements over the broadest range, from the fastest on-rates to the slowest off-rates. **(1)** Interactions with apparently similar affinities can have very different kinetic profiles. Resolution into component on- and off-rates can improve candidate selection. **(2)** Even interactions at the extremes of kinetic behavior, for example, with very slow off-rates, can be detected and differentiated with confidence.

Scope of performance

Analyze up to 384 samples in unattended runs

Biacore T200 supports the use of 96- and 384-well microplates and vials. The use of all four flow cells allows four interactions to be simultaneously monitored. The sample compartment of Biacore T200 can be cooled down to 4°C, enabling the analysis of sensitive samples in unattended runs of up to 48 h.

Predict in vivo behavior by studying interactions at physiological temperatures

By providing reliable data at physiological temperatures (Fig 6), Biacore T200 enables the *in vivo* behavior of therapeutics to be more confidently predicted.

An integrated buffer degasser prevents the formation of air bubbles at elevated temperatures, helping to ensure higher quality data. The integrated degasser also eliminates the need for buffer degassing before the run.

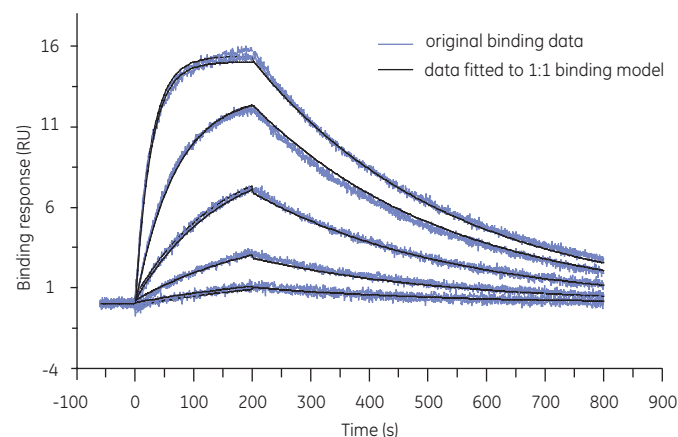


Fig 6. Biacore T200 delivers stable and reproducible data at 37°C.

Perform buffer scouting for fast assay development

With the built-in buffer selector, up to four buffers can be tested in a single run, accelerating assay development. For example, microenvironmental effects on binding properties can be studied in mechanistic and stability studies.

Recover samples for identification by mass spectrometry

Protein interaction analysis on Biacore T200 in combination with mass spectrometry provides the possibility to identify proteins on the basis of functional binding criteria. Sample recovery and digestion are supported by the software.

- Analytes recovered in a small volume, maximizing concentration
- Minimum carry-over from sample to recovered solution
- Deposition in vial containing digestion solution
- Entire recovery process predefined in software template

Software wizards and templates for ease-of-use

Biacore T200 software is suited to users at all levels of experience. Software wizards offer support throughout all assay steps from development to data interpretation, simplifying the entire process and reducing time-to-results (Fig 7).

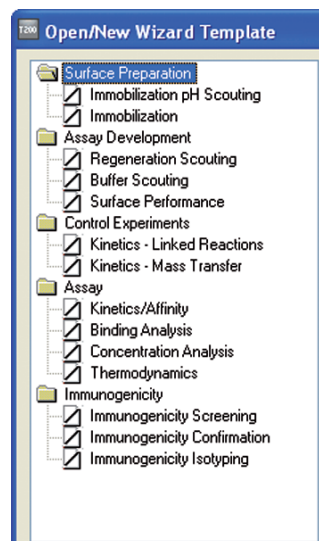


Fig 7. Software wizards for every interaction parameter including assay development.

Flexibility in assay design with Method Builder

Method Builder is intuitive, powerful software for flexible assay design. Predefined methods may be used directly or may be modified for specific applications (Fig 8). Alternatively, users may choose to develop novel, customized methods.

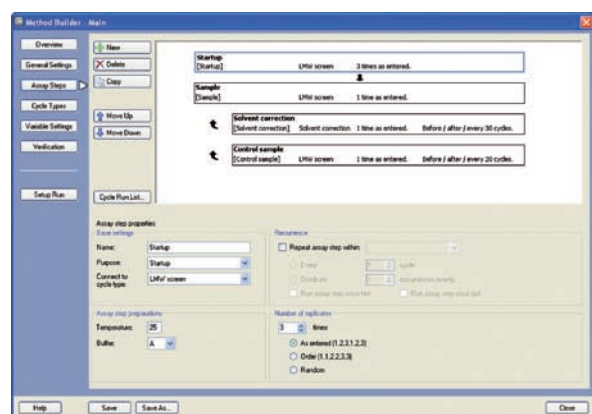


Fig 8. User-friendly Method Builder for flexible assay design.

Fast, simple kinetic analysis

Biacore T200 software offers a range of tools for confident and reliable kinetic analyses. These can be performed using a multicycle approach (using one sample concentration per cycle), or alternatively, using single-cycle kinetics. By eliminating the need for surface regeneration between injections, single-cycle kinetics simplifies analyses involving targets that are difficult to regenerate (Fig 9) and reduces assay development time.

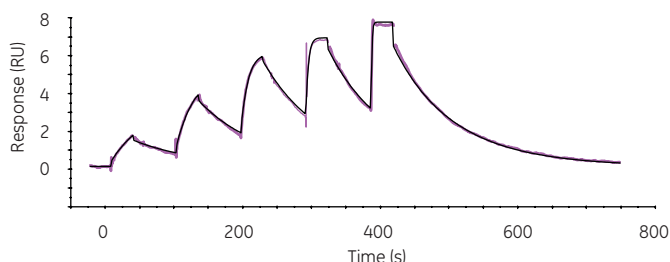


Fig 9. Using the single-cycle kinetics approach, samples are injected one after the other in the same cycle with no intervening regeneration steps. Here, a dilution series of a molecule with a relative molecular mass (M_r) of 312 was prepared at concentrations of 0.062, 0.185, 0.556, 1.667 and 5 μM , and sequentially injected over a sensor surface prepared with a protein (M_r 29 000) immobilized at 760 RU.

Biacore T200 Evaluation Software enables kinetic evaluation of interactions with a few simple clicks.

- Process multiple samples simultaneously to simplify the data analysis process and save time
- Fit the kinetic data using one of several available interaction models
- Automatically subtract reference and blank data

Simple QC tools for easy assessment of kinetic data

Automated QC tools analyze data fitting quality for the magnitude of kinetic constants, parameter uniqueness, bulk refractive index, and residuals, enabling the user to interpret results with confidence (Fig 10).

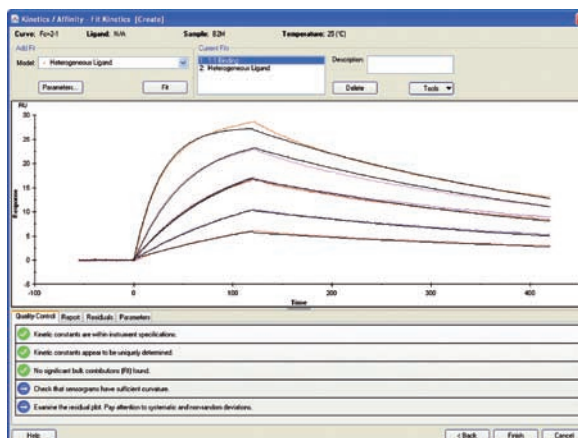


Fig 10. Integrated QC tools assess data quality to ensure valid data interpretation.

Evaluation tools for convenient data presentation, comparison, and selection

Results can be plotted and presented using a range of tools. The Kinetic Summary tool provides an overview of binding profiles such as on-off rate maps to aid comparison and selection (Fig 11 and 12).



Fig 11. Thumbnail plots with sorting possibilities for data comparison.

These maps provide an overview of kinetic and affinity properties via a logarithmic-scale plot of association rate constants (k_a) against dissociation rate constants (k_d). Points on any single diagonal represent interactions with similar affinities but possibly with very different kinetic profiles.

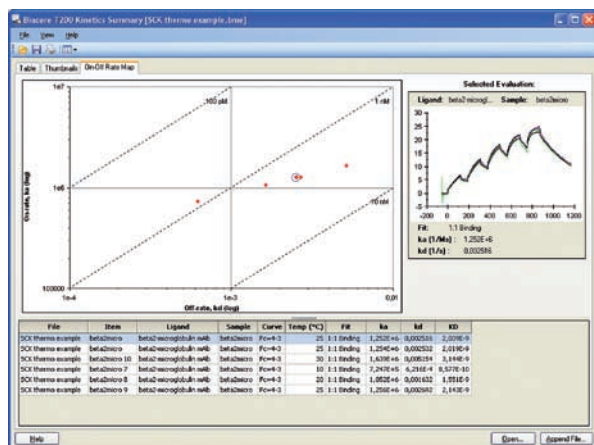
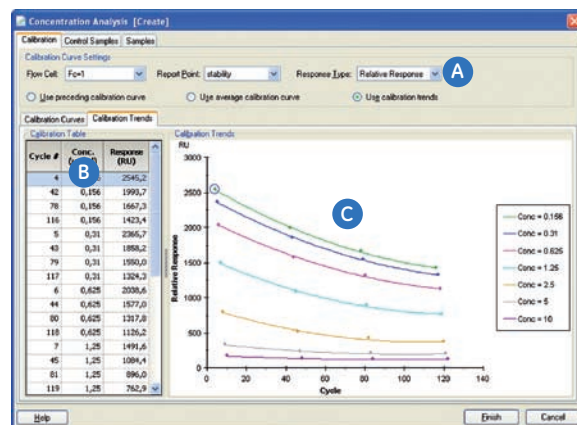


Fig 12. On-off-rate maps reveal how molecular interactions with identical affinities can have radically different kinetic profiles.

Confident concentration analysis – with or without standards

Software-supported direct binding and inhibition assays on Biacore T200 enable measurement of active concentration, and not just total protein (Fig 13). The precision and automation of the system generates highly reproducible data with CV typically below 5%.

- Generate highly reliable data from run to run by interpolation of repeated calibration curves
- Ensure rigorous QC by inclusion of control samples



Thermodynamic analysis for additional information about reaction mechanisms

Derive transition state thermodynamics from kinetic rate constants

Fully understanding molecular recognition by being able to predict binding energetics from the three-dimensional structure of protein complexes by thermodynamic analysis may well provide the basis for structure-based molecular design of drugs and engineered antibodies.

Dedicated software wizards, built-in buffer degassing, and temperature control in Biacore T200 make transition state thermodynamics easier than ever. Integration of rate constants measured across several temperatures into thermodynamic equations allows thermodynamic characterization of the transition state, revealing the forces driving the interaction (Fig 14).

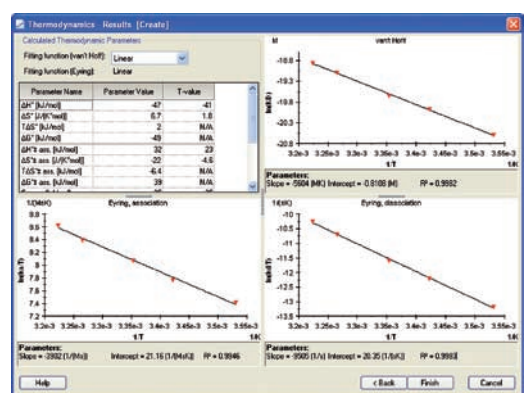


Fig 14. Automatic generation of Eyring and van't Hoff plots from kinetic data for calculation of thermodynamic parameters.

Dedicated support for immunogenicity testing

Biacore T200 software provides dedicated tools for immunogenicity testing for confident detection and characterization of anti-drug antibodies (ADA) during preclinical and clinical development. Biacore T200 may be integrated at several points throughout the entire immunogenicity workflow at screening, confirmation (elimination of false positives), and characterization (isotype determination, epitope mapping, neutralization capacity).

- Detect low affinity ADA, easily missed in endpoint assays due to losses during washing steps (Fig 15)
- Detect ADA even in the presence of drug, avoiding false negative results
- Comprehensive characterization of ADA with dedicated software tools

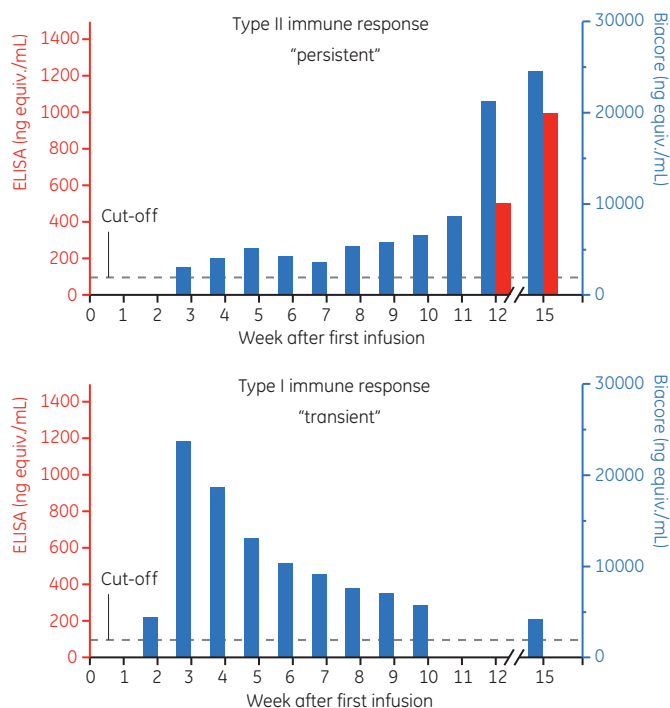


Fig 15. A comparison of data from a bridging ELISA assay and a Biacore assay in a Phase I study for a therapeutic antibody. Positive samples were detected much earlier in the Biacore assay and led to the implementation of Biacore as the preferred choice for immunogenicity screening. Data courtesy of Dr. Ulrich Kunz, Boehringer Ingelheim Pharma GmbH & Co. KG

Work in GxP-regulated environments

Optional software for GxP compliance

An optional GxP Package allows Biacore T200 to integrate seamlessly into GxP-regulated workflows. The package provides validated software supporting GLP/GCP/GMP and 21 CFR Part 11 compliance, and includes validation support. For full validation support during the lifetime of the system, the package can be supplemented with GE Healthcare Validation GxP Services.

Features in the GxP Package include:

- Data integrity – access control and enforced version handling
- User authorization levels – administrator, developer and user levels set access rights to software functions
- Published procedures for operational control – enables assay run and evaluation settings to be locked together in routine assays
- Audit trail – tracks record modifications and maintains complete version histories for published procedures
- Change Control Procedures (CCP), performed as required following hardware and software changes

Data can be exported both manually and automatically in Microsoft Excel (XLS) format as well as Extended Markup Language (XML) format. The software has been developed in accordance with an accepted development model to ensure adequate validation.

Biacore T200 specifications

Technical specifications and characteristics

Detection technology	Surface Plasmon Resonance (SPR) biosensor
Information provided	Kinetic and affinity data (K_D , k_a , and k_d), specificity, selectivity, concentration, and thermodynamic data
Data presentation	Result tables, result plots, and real-time monitoring of sensorgrams
Analysis time per cycle	Typically 2 to 15 min
Automation	48 h unattended operation
Sample type	LMW drug candidates to high molecular weight proteins (also DNA, RNA, polysaccharides, lipids, cells, and viruses) in various sample environments (e.g., in DMSO-containing buffers, plasma, and serum)
Required sample volume	Injection volume plus 20–50 μ l (application dependent)
Injection volume	2 to 350 μ l
Flow rate range	From 1 to 100 μ l/min
Flow cell volume	0.06 μ l
Flow cell height	40 μ m
Sample/reagent capacity	1 $\times$ 96- or 384-well microplate and up to 33 reagent vials
Analysis temperature range	4°C to 45°C (maximum 20°C below ambient temperature)
Sample storage	4°C to 45°C (maximum 15°C below ambient temperature)
Sample refractive index range	1.33 to 1.40
Buffer selector	Automatic switching between 4 buffers
In-line reference subtraction	Automatic

Minimum computer requirements

3.0 GHz processor
RAM > 1 GB free
CD-ROM drive
Hard disk drive > 2 GB free
Graphics resolution at least 1280 $\times$ 1024

Typical working ranges

Association rate constant (k_a)	Proteins: 10^3 to 3×10^9 $M^{-1}s^{-1}$ LMW molecules: 10^3 to 5×10^7 $M^{-1}s^{-1}$
Dissociation rate constant (k_d)	10^{-5} to 1 s^{-1}
Sample concentration	≥ 10 pM
Molecular weight detection	No lower limit for organic molecules
Number of flow cells	4
Baseline noise	Typically < 0.03 RU (RMS)
Baseline drift	Typically < 0.3 RU/min
Recovery specifications	1.5 μ l analyte recovery volume
Immobilized interactant consumption	Typically 0.03 to 3 μ g/flow cell
Dimensions (W $\times$ H $\times$ D)	600 $\times$ 615 $\times$ 690 mm
Net weight total	60 kg
Mains requirements	Processing Unit Autorange 100–240 VAC ($\pm 10\%$), 50–60 Hz, Class 1 equipment (protective earthing)
Power consumption	Processing Unit: max 6.3 A (at 100 VAC)

Data handling and storage

PC operating systems	Windows™ XP Professional edition, Windows Vista™ Business edition, and Windows 7 Professional edition
Interfacing	Possibilities for import of sample data and export of results (e.g., to and from LIMS)

Compliance

Safety standard	EU:EN61010-1(2001), EN61010-1-081 (2002), North America: UL61010A-1, and CAN/ CSA-C22.2 No.1010.1
EMC	EN 61326-1 (1997), Amendment A1 (1998), and Amendment A2 (2001)
GxP	Technical controls for 21 CFR Part 11 (ER) compliance (Biacore T200 GxP Package) Validation support, including Equipment Qualification (GE Healthcare Validation GxP Services)

Delivery: Biacore T200 includes a processing unit, control and evaluation software, and Windows operating system.

Optional: Biacore T200 GxP Package and GE Healthcare Validation GxP Services

On-site requirements: Contact your local GE Healthcare representative for the latest information regarding on-site requirements.

Ordering information

Processing unit*	Code no.
Biacore T200	28-9750-01

* Includes a processing unit, control and evaluation software, and Windows operating system.

Desktop package	Code no.
Computer, screen, and printer - 110 V	28-9227-26
Computer, screen, and printer - 220 V	28-9227-25

Optional packages†	Code no.
Biacore T200 GxP Package	28-9779-54
GE Healthcare Validation GxP Services	BR-2001-06

† For more information, contact your local GE Healthcare representative.

Related literature	Code no.
Biacore T200 Upgrade Kit, Data file	28-9794-14
Profiling inhibitor selectivity for PARP protein family members, Application note	28-9794-18
Outstanding sensitivity for confident label-free interaction analysis, White paper	28-9794-20
Biacore T200 system, Brochure	28-9794-12

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